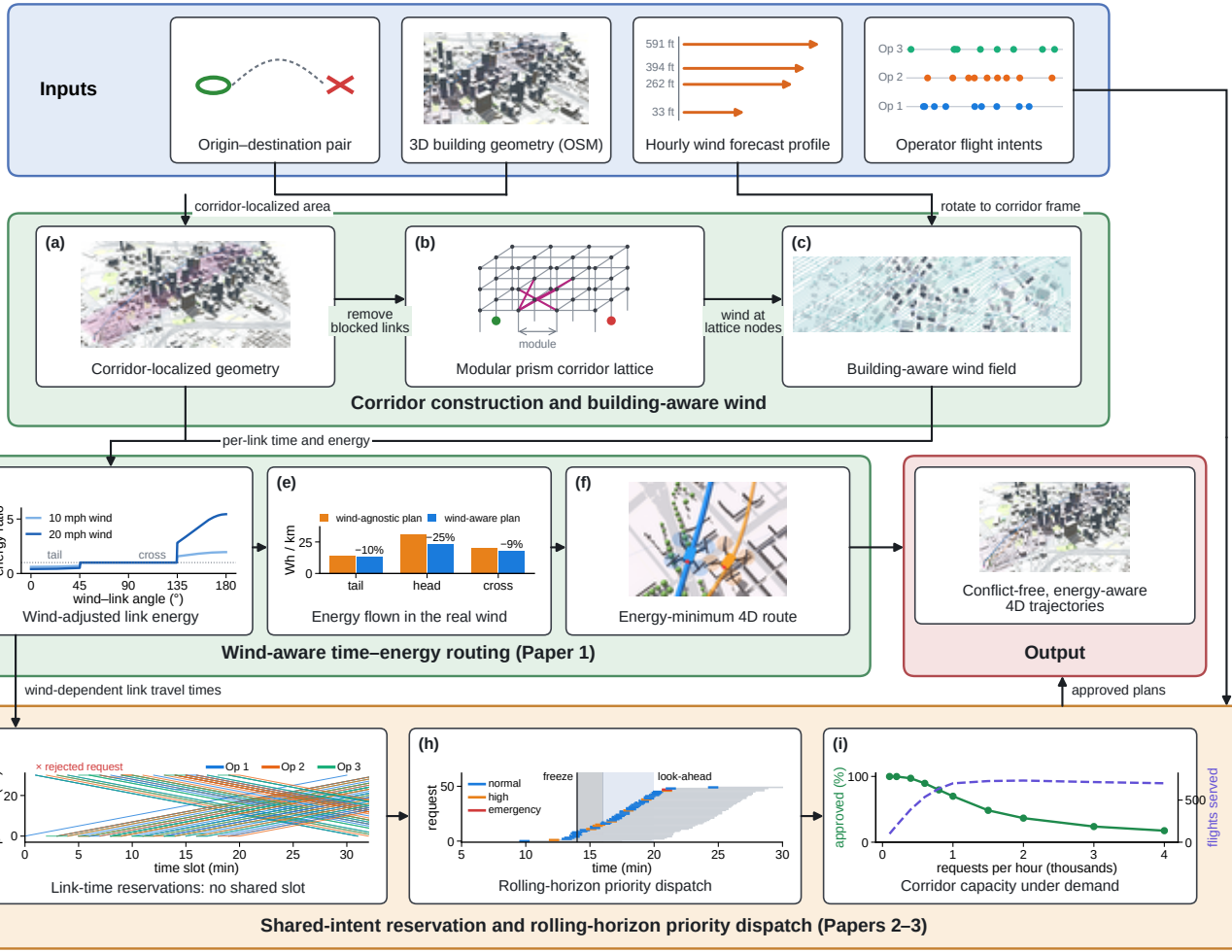

Inputs
Origin–destination pair
3D building geometry (OSM)
591 ft
394 ft
262 ft
33 ft
Hourly wind forecast profile
Op 3
Op 2
Op 1
Operator flight intents
corridor-localized area
rotate to corridor frame
flight intents
(a)
Corridor-localized geometry
remove blocked links
(b)
module
Modular prism corridor lattice
wind at lattice nodes
(c)
Building-aware wind field
Corridor construction and building-aware wind
per-link time and energy
(d)
energy ratio
10 mph wind
20 mph wind
tail
cross
wind–link angle (°)
Wind-adjusted link energy
(e)
wind-agnostic plan
wind-aware plan
Wh / km
−10%
−25%
−9%
tail
head
cross
Energy flown in the real wind
(f)
Energy-minimum 4D route
Wind-aware time–energy routing (Paper 1)
Conflict-free, energy-aware 4D trajectories
Output
wind-dependent link travel times
approved plans
(g)
× rejected request
Op 1
Op 2
Op 3
position (kft)
time slot (min)
Link-time reservations: no shared slot
(h)
normal
high
emergency
freeze
look-ahead
request
time (min)
Rolling-horizon priority dispatch
(i)
approved (%)
flights served
requests per hour (thousands)
Corridor capacity under demand
Shared-intent reservation and rolling-horizon priority dispatch (Papers 2–3)